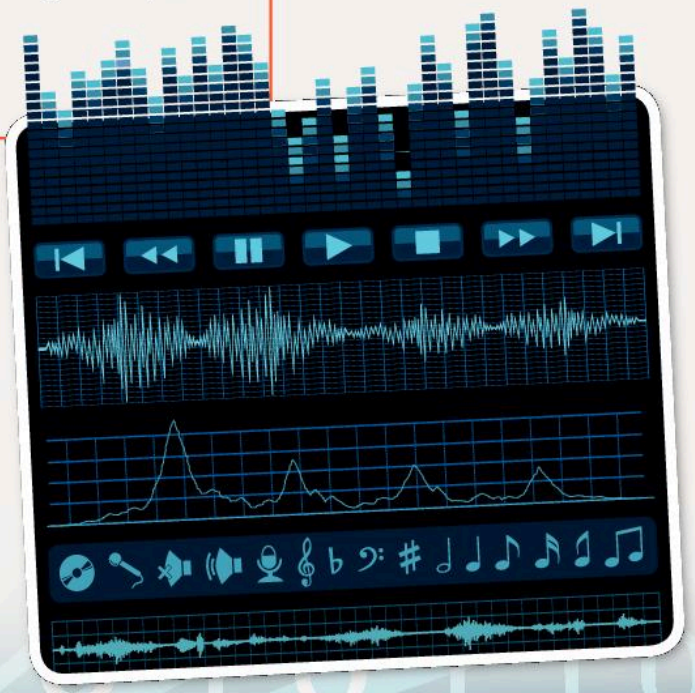
# 1 The nature of technology: Storing digital data

## You will learn

- how text, images and audio can be stored as digital data by a computer
- how to convert between binary and decimal numbers
- how to add binary numbers.



A computer stores data in digital files. A digital file only contains the characters zero and one. In this unit you will learn how text, photographs and images can be turned into digital data so that they can be stored on a computer. You will convert everyday decimal numbers into digital data. You will use codes that will help you understand how computers store text as numbers. You will create simple images and convert them into digital data in the same way as a computer. You will discover how digital sound and video are created.



### Talk about...

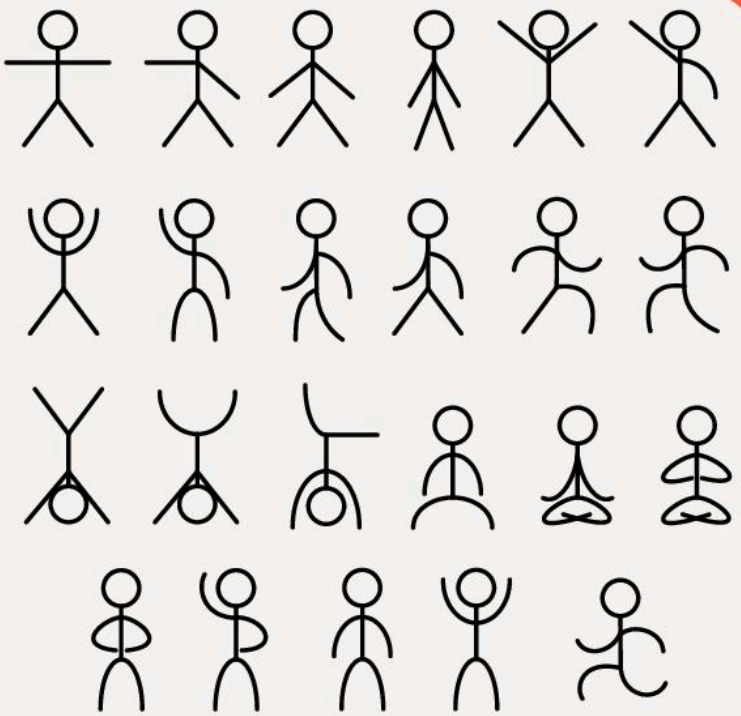
More of our personal data is being stored on the internet than ever before. We choose to store some of that information ourselves on social media sites. Governments, banks and online retailers save information about us online. Do you worry about your data being stored on the internet? Is your data safe?

**Curriculum reference:** Describe how different types of data can be represented in binary digital form; Convert between decimal and binary integers; Perform simple binary additions

## Unplugged

Video and animation are created by showing a series of still images very quickly, one after the other. Create your own animation using a flick book. You will need 15 to 20 strips of paper stapled together. Draw a simple image on the first strip then change it slightly on each of the following strips.

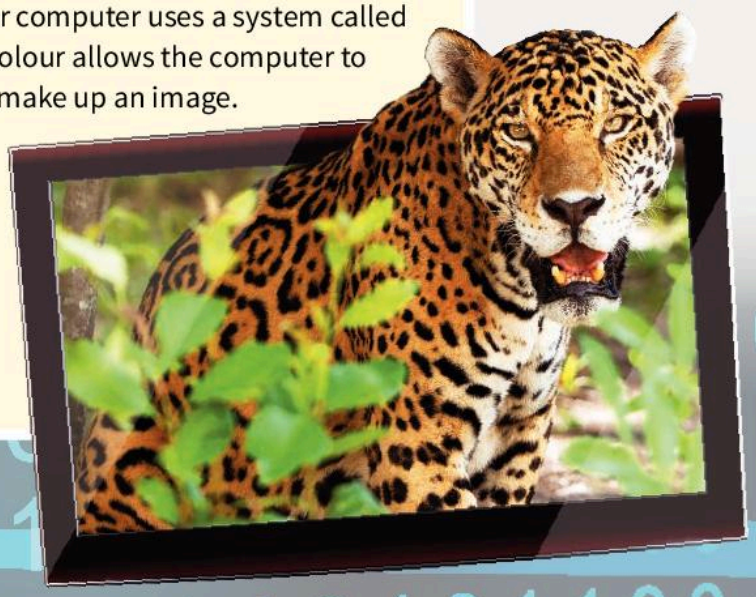
An easy animation to draw is a bouncing ball. If you are feeling more creative, draw a dancing stick character. When you have finished, flick through the paper strips to animate your drawing.



## Did you know?

Computers and other digital devices such as televisions display realistic-looking photographs and video images. Your computer uses a system called true colour to create realistic images. True colour allows the computer to store information about all the shades that make up an image.

True colour allows a computer to use more than 17 million colours in an image. That is more colours than most humans can see. Storing the information about a single true colour takes up the same amount of space as the computer uses to store the word 'red'.



binary digital data bit
byte code ASCII media
pixel true colour sampling
Unicode

# 1.1 Digital data

## In this lesson

You will learn:

- what digital data is
- that computers store digital data as binary numbers
- how digital data is used to store numbers, media and instructions.

**Spiral back**

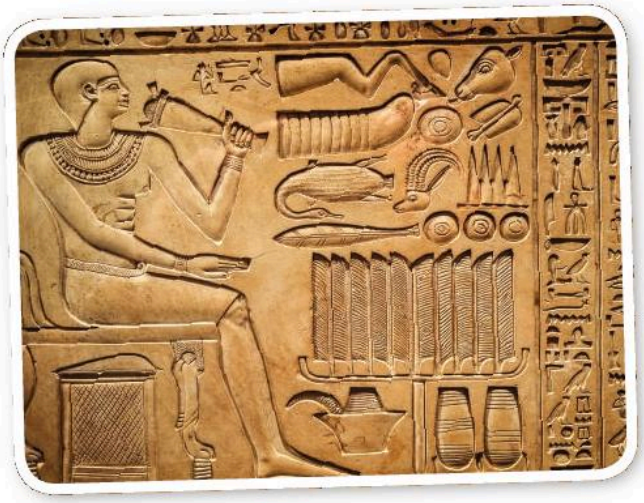
In Student Book 4 you learned about the different types of computer we can use to help us work and in our leisure time. All computers are digital devices – they store and process digital data. In this unit you will learn what digital data is and how computers use it to store text, images and sound.

## Storing data

Humans store data – lots of data. We need to store data so that we can use it again when we need it.

People have stored data since ancient times. Early people stored data about important events by painting images on cave walls and by carving images and hieroglyphics into stone. Later people recorded their history and scientific achievements by writing on scrolls and parchment.







Throughout history, people developed different ways to store data. They developed printing so that they could store text and images in books. They developed vinyl records, tapes and then CDs to store and play music.

In the modern world, people use computers to store and process data. Most of the data that people use today is stored in a format that computers can use.

## What is digital data?



When you communicate in English, you use letters and numbers. You use 26 letters and 10 digits, zero to nine. You also use punctuation characters such as the comma and full stop. You combine these characters to make words and sentences.

A computer stores data using only digits. Data that is stored using digits is called **digital data**. The computer uses only two digits: the numbers zero (0) and one (1).

Every file stored on a computer is made up of zeros and ones. A computer can use digital data files to store text, images, video or audio.

## What does a computer use digital data for?

The 1s and 0s inside the computer can be used to store:

- yes and no (or true and false)
- numbers
- instructions that tell the computer what to do
- other digital content such as text, images and sounds.



## Binary numbers

When you do use maths to solve day-to-day problems, you use the decimal number system. The decimal system has 10 different digits: the digits 0 to 9. The 'dec' in the word 'decimal' means 10. Some people think that we started to use the decimal system because we use our 10 fingers to count with.

The number system that a computer uses has two different digits: 0 and 1. This is called the **binary** number system. The 'bi' in the word 'binary' means two.

A computer uses binary to store numbers.

Binary numbers are used in calculations in the same way as decimal values. The table shows some example values written as both decimal and binary numbers. They look different but they mean the same thing.

**Decimal and binary numbers**

| Decimal | Binary |
|---|---|
| 1 | 1 |
| 8 | 1000 |
| 18 | 10010 |
| 100 | 1100100 |

### Activity

Look at the table of binary and decimal numbers. Describe any differences and similarities between decimal numbers and binary numbers that you notice.

## Storing text using binary numbers

A computer uses binary to store media. Binary can store text, images, sound and even video. When a computer uses binary to store media it uses codes.

To a computer the word 'Hello' looks like this: 01001000 01100101 01101100 01101100 01101111.

Each set of eight digits is a code for a letter. The code for 'H' is 01001000. The code for 'l' is 01101100. The code for 'l' is used twice in the binary word because there are two 'l's in 'Hello'.

| 01001000 | 01100101 | 01101100 | 01101100 | 01101111 |
|---|---|---|---|---|
| **H** | **e** | **l** | **l** | **o** |

In an image, codes are used to represent colours. In a music file, codes can be used to represent different instruments. Complex photographs and music files are all stored as zeros and ones.

### Activity

Using the binary number codes in the word 'Hello', translate the word below into English.

01001000 01101111 01101100 01100101

## Storing instructions using binary numbers

The instructions in computer programs are stored in binary. In Unit 3 you will write computer programs. The instructions are written using human alphabets and symbols. The instructions in a program must be converted into binary so that the computer can store and use them. Each binary instruction tells the computer to do a simple task.

### Activity

Your task is to program a robot to find its way through a maze. There are just four simple instructions that you can give to the robot. These instructions tell the robot which direction to move:

- one step left
- one step right
- one step up
- one step down.

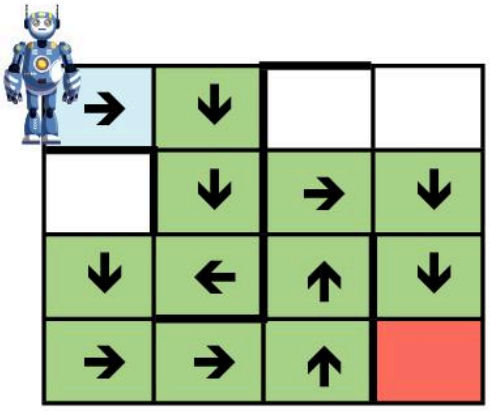
The binary codes for these instructions are shown in the table on the right.

| Action | Code |
|---|---|
| One step left | 00 |
| One step right | 01 |
| One step up | 10 |
| One step down | 10 |

Use the binary codes to write the program for the robot. Your program should follow the arrows shown in the diagram below to follow the green path through the maze from the blue square to the red square. The first five instructions are:

01, 11, 11, 00, 11

These instructions mean: 'Move one step right. Move one step down. Move one step down. Move one step left. Move one step down.'



Complete the list of instructions to guide the robot to the red square.

## Why do computers use digital data?

Computers are powered by microprocessors. The microprocessor is the 'brain' of a computer. A microprocessor is made up of millions of tiny electronic switches. The switches in a microprocessor are like any other switch. A switch can either be on or off.

A microprocessor is called a **digital** device because it can only understand the two switch positions – on and off. The on and off positions can be shown as 1 and 0 in binary. You have already learned that digital data is made up of 1s and 0s. That is why a digital microprocessor can read digital data.

### Extra challenge

Think about what you have learned in your computing course over the last year. List examples of activities and assignments where you have stored data as values, media and instructions to the computer.

### Test

1. Write down the two digits used in the binary number system.
2. Write down the eight extra digits used in the decimal system that are not used in binary.
3. In your own words, explain what digital data is.
4. Describe three things a computer stores as binary code.

# 1.2 Reading binary numbers

## In this lesson

You will learn:

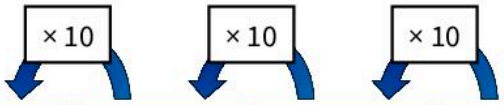
- about bits and bytes
- how to convert binary numbers into decimal numbers
- the meaning of base 2 and base 10.

## Understanding binary

### Base 10 and base 2 numbers

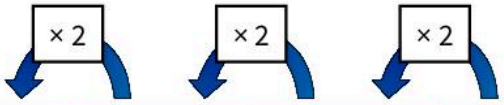
In the last lesson you learned that a computer must store everything it processes as digital data. You learned that you can use the binary number system to understand what digital data looks like. You also compared some binary numbers to decimal numbers.

The decimal number system uses 10 digits (0 to 9). The value of each column in a decimal number is 10 times greater than the previous column. Another name for the decimal system is **base 10**.




| 1000s | 100s | 10s | Units |
|---|---|---|---|
| 2 | 5 | 4 | 3 |

The binary number system uses two digits (0 and 1). The value of each column in a binary number is 2 times greater than the previous column. Another name for the binary system is **base 2**.




| 8s | 4s | 2s | Units |
|---|---|---|---|
| 1 | 1 | 0 | 1 |

### Activity

Other number systems are sometimes used in computing. One system is octal. Octal is the base 8 number system. Use the information you have learned about base 2 and base 10 to answer these questions:

- How many digits does the base 8 system use?
- What are those digits?
- What are the values of the first four columns in the base 8 number system? Draw a table to show your answer.

## How to read binary numbers

You can use your knowledge of base 2 to read binary numbers. The easiest way to understand a binary number is to convert the binary number into a decimal number. You use the decimal system every day, so it is much easier to understand.

The binary number shown in the table above is 1101. Here is an easy way to convert the number into decimal.

**1** Draw a table like the ones in the example on the previous page. It must have enough columns to hold the binary number you want to convert.

2 In the first row of the table, write the value of each column. Start with units in the right-hand column then multiply by 2 each time you move from right to left.

| 8s | 4s | 2s | Units |
|---|---|---|---|
| | | | |

**3** Write the number you want to convert in the second row of your table

| 8s | 4s | 2s | Units |
|---|---|---|---|
| 1 | 1 | 0 | 1 |

**4** Multiply each digit in the number you want to convert by the column value.

$1 \times 8 = \mathbf{8}$ $1 \times 4 = \mathbf{4}$ $0 \times 2 = \mathbf{0}$ $1 \times 1 = \mathbf{1}$

**5** Add the results together. The total is the value of the binary number as a decimal number.

$8 + 4 + 0 + 1 = \mathbf{13}$

1101 in binary is 13 in decimal.

### Activity

Convert these binary numbers into decimal numbers.

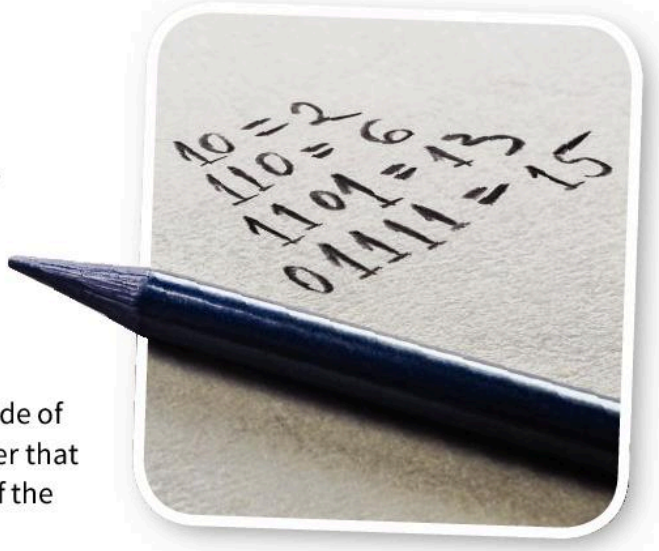
**a** 0111

**b** 1001

**c** 11001

**d** 111001

You will need to add more columns to the left-hand side of your table for the numbers in parts c and d. Remember that the value of each column must be 2 times the value of the column to the right.



## Bits and bytes

Each digit in a binary number is called a **bit**. There are four bits in the binary number 1101. The word bit is short for '**b**inary dig**it**' – the first letter of 'binary' combined with the last two letters of 'digit'.

A bit isn't very useful on its own. It can only store one of two values: 0 or 1. To make binary more useful, computers group bits together. Eight bits grouped together is called a **byte**. Here are some examples of data stored in a byte: 11111111, 00000000, 00110101.

When you write a byte you must show all eight digits in the number, even if you have to start the number with zeros. The value zero as a byte is 00000000.

## The language of numbers

'Ten' and 'eleven' are examples of names of numbers in the decimal system. Every number in the decimal system has a name. Numbers are not given names in binary. The binary number 11 is called 'one-one'. The decimal number eleven and the binary number 11 are different numbers.

**Eleven (decimal)**



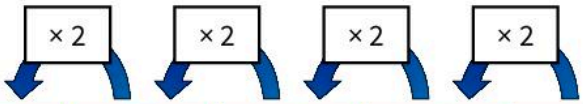
**11 (binary)**



### Activity

Copy the table below and complete the column headings.

Then use the table to convert the byte 01100110 into decimal.




| ? | ? | ? | ? | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |

## A binary to decimal shortcut

Base 2 is very easy to convert into decimal because it only uses two digits: 0 and 1. When you feel confident using the method of converting binary into decimal you have learned in this lesson, you can try this shortcut:

Use the byte table you created in the last activity. Start at the right of the binary number. Look at each digit in turn. Add together the place value of every column that contains a 1. With practice, you will soon learn the column heading values and be able to convert binary numbers in your head. Try this shortcut method when you complete the next activity.

### Activity

The table below contains the values 0 to 9 written as binary numbers. The numbers are in random order. Rewrite the list of numbers so that they are in order from zero to nine.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0011 | 0010 | 1000 | 0000 | 0110 | 0111 | 0101 | 0001 | 0100 | 1001 |

### Extra challenge

Work with a partner or in two small teams.

Write each of the binary numbers from the table in the last activity on a separate card. Mix the cards up and lay them out facing your partner. Now set your partner some challenges.

- Point to a card and ask your partner to tell you the decimal value of the card.
- Pick three cards at random and ask your partner to place them in numerical order.
- Ask your partner to pick two cards that add up to the decimal value 6 (or choose another value).




### Test

1. Convert the binary number 1001 into decimal.
2. a What is a byte?
   b Convert the byte 11001011 into decimal.
3. Look at this binary number: 00100010. Explain why the 1 in column 6 has a different value to the 1 in column 2.
4. Explain what base 2 means.

### Explore more

Teach a friend or family member how to read binary numbers. Challenge them to a contest.

# 1.3 Adding in binary

## In this lesson

You will learn:

- how to do simple addition in binary
- what overflow means when adding in binary.

When digital data is stored as a value it can be used in calculations. For example, you have used spreadsheets in this course. When you enter a formula such as =A3+B3 in a spreadsheet cell your computer will carry out binary addition. You have also learned programming in this course. When your computer moves a sprite on screen it uses binary addition to calculate a new position.

In this lesson you will learn how to do simple addition using binary.

## Simple addition in decimal

To do addition in binary you use the same method as addition in decimal. It will help to look at an example of how addition works in decimal before moving on to binary addition.

If you lay out the addition in a table like the one on the right, it is easier to follow what happens when you add the numbers. You use the first two rows in the table for the numbers you will add together. You use the row at the bottom to record the sum. You use the shaded row to hold any values that you need to carry.

| | | | |
|---|---|---|---|
| Number 1 | | | |
| Number 2 | | | |
| Carry | | | |
| **Sum** | | | |

### Example: decimal addition

In this example you will add together the numbers 262 and 174.

When you do an addition, you add together the numbers in each column from right to left and record the sum.

**Step 1:** Add the Units column: Add 2 and 4 together. Think of the sum as being '06'. This will help when you do binary addition later.

Write '06' in the table. Enter the 0 in the carry row of the 10s column. Enter the 6 in the sum row of the Units column.

| | 100s | 10s | Units |
|---|---|---|---|
| Number 1 | 2 | 6 | 2 |
| Number 2 | 1 | 7 | 4 |
| Carry | | 0 | |
| **Sum** | | | **6** |

**Step 2:** Add the 10s column: The digits in this column add up to 13. Enter the 1 in the carry row of the 100s column. Enter the 3 in the sum row of the 10s column.

| | 100s | 10s | Units |
|---|---|---|---|
| Number 1 | 2 | 6 | 2 |
| Number 2 | 1 | 7 | 4 |
| Carry | 1 | 0 | |
| **Sum** | | **3** | **6** |

**Step 3:** Add the 100s column: The digits in this column add up to 04. There is nothing to carry. Enter the 4 in the sum row of the 100s column. This completes the addition.

| | 100s | 10s | Units |
|---|---|---|---|
| Number 1 | 2 | 6 | 2 |
| Number 2 | 1 | 7 | 4 |
| Carry | 1 | 0 | |
| **Sum** | **4** | **3** | **6** |

262 + 174 = 436.

## Activity

Draw a copy of the table used in the previous example. Use it to add 729 and 252.

## Adding in binary

You use the same method to add two binary numbers together. Adding in binary seems more difficult because you are not so familiar with binary numbers. There are four rules that will help you to do binary addition.

**Rule 1:** 0 + 0 = **00**

| | | |
|---|---|---|
| Number 1 | | 0 |
| Number 2 | | 0 |
| Carry | **0** | **0** |
| **Sum** | | **0** |

**Rule 2:** 0 + 1 = **01**

| | | |
|---|---|---|
| Number 1 | | 0 |
| Number 2 | | 1 |
| Carry | **0** | **0** |
| **Sum** | | **1** |

**Rule 3:** 1 + 1 = **10**

| | | |
|---|---|---|
| Number 1 | | 1 |
| Number 2 | | 1 |
| Carry | **1** | **0** |
| **Sum** | | **0** |

**Rule 4:** 1 + 1 + 1 = **11**

| | | |
|---|---|---|
| Number 1 | | 1 |
| Number 2 | | 1 |
| Carry | **1** | **1** |
| **Sum** | | **1** |

You can use these rules to add any two binary numbers. Write the two numbers one above the other so the bits line up. Start with the bits in the Units column (on the right). Look down the column. What bits do you see? It will be one of these possibilities:

- 0 + 0
- 1 + 0
- 1 + 1

The rules of binary addition will tell you the answer. Write in the answer and any carry bit.

Now go to the next column (the 2s column). Look down the column, including any carry bit. What bits do you see? The same possibilities as before. Because of the carry bit there is another possibility:

- 1 + 1 + 1

Write in the answer plus any carry bit. Do the same for every column until you have added every column.

You will see an example on the next page.

## Example: binary addition

In this example you will add the binary numbers 0011 and 1011. You will use the four rules to help you with your binary addition.

**Step 1:** **Add the Units column:** Number 1 and Number 2 both have a 1 in the Units column. Rule 3 says 1 + 1 = **10**. Enter the 1 in the carry row of the 2s column. Enter the 0 in the sum row of the Units column.

**Step 2:** **Add the 2s column:** Number 1 and Number 2 both have a 1 in the 2s column. There is also a 1 in the carry row. Rule 4 says 1 + 1 + 1 = **11**. Enter the first 1 in the carry row of the 4s column. Enter the second 1 in the sum row of the 2s column.

**Step 3:** **Add the 4s column:** Number 1 and Number 2 both have a 0 in the 4s column. There is a 1 in the carry row. Rule 2 says 0 + 1 = **01**. Enter 0 in the carry row of the 8s column. Enter 1 in the sum row of the 4s column.

**Step 4:** **Add the 8s column:** Rule 2 says 0 + 1 = **01**. You don't need to enter the 0 because there are no more columns. Enter 1 in the sum row of the 8s column.

**Step 1** Rule 3: 1 + 1 = 10

| | 8s | 4s | 2s | Units |
|---|---|---|---|---|
| Number 1 | 0 | 0 | 1 | 1 |
| Number 2 | 1 | 0 | 1 | 1 |
| Carry | | | 1 | |
| **Sum** | | | | 0 |

**Step 2** Rule 4: 1 + 1 + 1 = 11

| | 8s | 4s | 2s | Units |
|---|---|---|---|---|
| Number 1 | 0 | 0 | 1 | 1 |
| Number 2 | 1 | 0 | 1 | 1 |
| Carry | | 1 | 1 | |
| **Sum** | | | 1 | 0 |

**Step 3** Rule 2: 0 + 1 = 01

| | 8s | 4s | 2s | Units |
|---|---|---|---|---|
| Number 1 | 0 | 0 | 1 | 1 |
| Number 2 | 1 | 0 | 1 | 1 |
| Carry | 0 | 1 | 1 | |
| **Sum** | | 1 | 1 | 0 |

**Step 4** Rule 2: 0 + 1 = 01

| | 8s | 4s | 2s | Units |
|---|---|---|---|---|
| Number 1 | 0 | 0 | 1 | 1 |
| Number 2 | 1 | 0 | 1 | 1 |
| Carry | 0 | 1 | 1 | |
| **Sum** | 1 | 1 | 1 | 0 |

## Activity

A student wanted to add 1010 + 0010. They put the numbers into the addition table.

| | 8s | 4s | 2s | Units |
|---|---|---|---|---|
| Number 1 | 1 | 0 | 1 | 0 |
| Number 2 | 0 | 0 | 1 | 0 |
| Carry | | | | |
| **Sum** | | | | |

Copy the table. Use the rules of binary addition to complete the table and find the sum.

Now use the same method to add 0011 + 0111.

## Adding bytes

In the example of binary addition in this lesson, you added together two four-bit binary numbers. The method works for binary numbers with any number of bits. You learned in Lesson 1.2 that computers use bytes to store and process data. A byte is 8 bits long. To add bytes together, extend the table used in the example so that it holds 8 bits.

### Activity

A student wanted to add two 8-bit binary numbers:

00110111 + 01001010

They extended the addition table to eight columns and put the two numbers into the table.

| | 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|---|
| Number 1 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 |
| Number 2 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 |
| Carry | | | | | | | | |
| **Sum** | | | | | | | | |

Copy the table. Use the rules of binary addition to complete the table and find the sum.

Now use the same method to add 01011001 + 00001111.

### Extra challenge

In this section you carried out a lot of binary calculations. You can check the results for yourself:

- Look at the two binary numbers you added together. Convert them both into decimal numbers.
- Look at the binary number that is the result of the addition. Convert that into a decimal number.

Now you have three decimal numbers. Add the first two numbers together. The result should equal your third number. If it all matches then your binary addition is correct.

Check all your binary addition sums using this method.

### Test

1. What is 1 + 1 in binary?
2. What are the four rules of binary addition?
3. Complete this binary addition: 00101101 + 00100101
4. Convert all the numbers in the sum in question 3 to decimal.

# 1.4 Decimal to binary

## In this lesson

You will learn:

- how to convert between decimal and binary
- what an overflow error is.

## Convert decimal to binary

When a decimal value is stored in the computer it must be converted to binary. In this lesson you will learn to convert decimal numbers into binary.

## Place values

The 1s of a binary number have different values depending on their place in the number. You have learned to show those values as column headings.

Here is an example. It shows the binary number 00100100. The digits have been placed a table with all the column headings.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |

You can find the value of each 1 by looking at the column headings. The first 1 is in the 32s column. It has the value 32. The second 1 is in the 4s column. It has the value 4. You can find the total value of the number by adding up the place values. In this case, 32 + 4 = 36.

So the binary number 00100100 has the decimal value 36.

## Convert decimal to binary

You can use the same number table to convert any decimal number to binary.

- Start at the left of the table.
- Subtract the place values from the decimal number (without going below 0).
- Whenever you subtract a value put a 1 into the table.
- Put 0 in all the other columns.

### Worked example 1

Here is a worked example. You will convert the decimal number 20 into binary.

First draw the grid.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

Start at the left. The first value is 128. That is too big to subtract from 20. So put a 0. Count along until you find a value that you can subtract. That is 16.

Put 1 into the 16s column.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | | | | |

Subtract 16 from 20, and there is 4 left. Put 1 into the 4s column.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | | |

You have now got 0 left so fill up the rest of the table with 0s.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |

So 20 in decimal is 00010100 in binary.

## Worked example 2

Convert the decimal number 161 into binary.

Counting from the left we can subtract values like this:

165 – 128 = 37

37 – 32 = 5

5 – 4 = 1

1 – 1 = 0

The numbers subtracted are 128, 32, 4 and 1. Put 1s into these columns.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 1 | | 1 | | | 1 | | 1 |

Put 0s in the other columns.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |

So 161 in decimal is 10100101 in binary.

## Activity

Convert the following decimal numbers to binary. Show your working.

**a** 32

**b** 80

**c** 69

**d** 133

## The largest number

In this unit you have learned to make numbers with up to 8 bits. That is 1 byte.

- The smallest number you can make with 8 bits is 00000000. That has the decimal value 0.
- The biggest number you can make with 8 bits is 11111111.

What is the decimal value of this number? You can find out by putting it into the binary grid.

| 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255

So the biggest decimal number you can make with 8 bits has the value 255.

## What about other numbers?

In real life the computer needs to store numbers outside of this range.
For example:

- numbers bigger than 255
- negative numbers (smaller than 0)
- fractions and decimal values, such as 4.5.

The computer needs to use more than 8 bits (one byte) to store these numbers.
In this unit you will only work with numbers from 0 to 255.

### Activity

Which of these decimal numbers **cannot** be turned into a binary number using 8 bits?

**a** 99

**b** 222

**c** 260

**d** 499

## Overflow error

If you try to store a number bigger than 255 using 8 bits then you will get an error. This is called an overflow error. In Lesson 1.3 you learned about binary addition. Sometimes the result of a binary addition will be a number that is bigger than 255. In this case you will get an overflow error.

## Worked example

Add the binary numbers 10101000 + 01100001.

Put them into the addition grid. Carry out the sum using the rules of binary addition.

| | 128s | 64s | 32s | 16s | 8s | 4s | 2s | Units |
|---|---|---|---|---|---|---|---|---|
| Number 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| Number 2 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |
| Carry | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| **Sum** | **0** | **0** | **0** | **0** | **1** | **0** | **0** | **1** |

The final column is the 128s column. There are two 1s in this column. The rules of binary addition tell us that 1 + 1 = 0 and carry 1. But there is nowhere to put the carry value. There are no more columns.

This causes an overflow error. The result is shown as 00001001. That is **not** the right answer to the sum.

## Extra challenge

Add the two binary numbers 11001011 + 00111111. Show the overflow error.

## Test

1. Convert the decimal number 67 to binary.
2. What is the largest decimal number that can be shown using 8 bits?
3. Complete the following binary addition, showing the overflow error.

   01011100 + 11001000
4. Convert the following decimal numbers into binary and show that adding them together causes an overflow error.

   150 + 120

# 1.5 Digital text and numbers

## In this lesson

You will learn:

- how numbers are stored as digital data
- how text is stored as digital data.

## What is a value?

You use numbers every day. Some of those numbers are used in calculations. For example, you add or multiply the numbers together. This type of number is called a **value**.

Other numbers are not values. A phone number is not a value. You don't add phone numbers together or subtract one phone number from another.

When these two types of number are stored as digital data they are stored in different ways.

### Activity

Discuss with your group which of these numbers are values and which are not values. Report your findings to the class.

**a** your house number

**b** the number of marks you get in a test

**c** your age

**d** the number of points your team scores in a basketball game.

## Other types of content

Number values are stored inside the computer using the binary number system. But what about other types of content? For example:

- text
- pictures
- sounds.

All of these types of content must be turned into numbers. Then the computer can store these numbers using the binary number system.

In the rest of this unit you will learn how the computer stores text, images and sounds as number values.

## Storing text using ASCII code

Text characters include letters of the alphabet, punctuation marks, and other characters that you can type on a standard keyboard, such as spaces and maths symbols.

The computer uses a number code to represent these text characters. There is a common code used by almost all computers. It is called ASCII (ask-ee). Here are number codes for the letters a to g in lower case. Different codes are used for upper-case letters.

| Character | Number code |
|---|---|
| a | 97 |
| b | 98 |
| c | 99 |
| d | 100 |
| e | 101 |
| f | 102 |
| g | 103 |

## How the computer uses ASCII code

When you press a key on the keyboard it sends a signal to the processor. The signal varies according to which character you selected on the keyboard.

When the processor receives the signal it stores the character using ASCII code. It stores the binary number. So, for example, the letter 'a' is stored as the binary number 01100001. The letter 'b' is stored as binary number 01100010, and so on.

Each ASCII code takes up exactly 1 byte (8 bits).

### Activity

Make an ASCII table.

**a** Copy the ASCII table in your book.

**b** Extend the table to show the whole alphabet from a to z.

**c** Add an extra column to the table to show the binary number for each character.

Use the ASCII table.

- Write your first name in ASCII using decimal values.
- Write your first name in ASCII using binary values.

## Storing other characters

You have used ASCII to represent lower-case letters from a to z. ASCII can also be used to store upper-case letters and other keyboard symbols such as punctuation marks.

The next table shows the ASCII codes for some common keyboard characters.

| Keyboard character | ASCII code (decimal) | ASCII code (binary) |
|---|---|---|
| **Space** | 32 | |
| **Comma** | 44 | |
| **Full stop** | 46 | |

### Activity

**a** Copy and complete the ASCII table to show the binary codes for the three keyboard characters.

**b** Here is a message in decimal ASCII. What does it say?

101 118 101 114 121 032 099 111 109 112 117 116 101 114 032 117 115 101 115 032 116 104 105 115 032 099 111 100 101

**c** Here is a message in binary ASCII. What does it say?

01111001 01101111 01110101 00100000 01110111 01101001 01101110

## Numbers in ASCII

The digits from 0 to 9 also have ASCII codes. The ASCII code for a digit is not the same as its number value.

| Keyboard character | ASCII code (decimal) | ASCII code (binary) |
|---|---|---|
| **0** | 48 | |
| **1** | 49 | |
| **2** | 50 | |
| **3** | 51 | |
| **4** | | |
| **5** | | |
| **6** | | |
| **7** | | |
| **8** | | |
| **9** | | |

The computer can store a number using ASCII or as a number value. Different types of software store numbers in different ways.

- If you type the number 39 into a word-processed document, the computer will store the ASCII codes for 3 and 9. You cannot do a calculation using a word processor.
- If you type the number 39 into a spreadsheet, the computer will store the number value 39. You can do calculations using a spreadsheet.

## Activity

Copy and complete the number table to show the ASCII codes for all the digits from 0 to 9.

## A problem with ASCII

There are only 256 characters in ASCII. ASCII was originally created to convert from English into binary. However, people around the world need to use computers in their own language.

An improved code called **Unicode** was invented in 1991. Around 110,000 characters are available in Unicode, including the ASCII codes. Unicode has codes for characters in Arabic, Mandarin and Japanese, as well as many other languages.

- ASCII uses a single byte to store the code for each character. The maximum number of characters a single byte can hold is 256.
- Unicode uses more than one byte to store characters. Two bytes joined together can hold 65,000 characters. Three bytes joined can hold nearly 17 million characters.

## Activity

Work with a partner. Write a short message in ASCII with about 10 characters and no punctuation. Give your message to your partner to decode. Work together to check you have coded and decoded each message correctly.

## Extra challenge

Search the web to find a full ASCII character code table. Make sure that the table includes the binary codes. Look at the ASCII codes for lower-case and upper-case letters. Explain how they are different.

## Test

1 What does 24 look like when it is stored as:
   a a value in binary?
   b ASCII?
2 When you type a character on your keyboard, what data is sent to the computer?
3 Why is ASCII limited to 256 characters?
4 What advantages are there to using Unicode instead of ASCII?



# 1.6 Sound and vision

## In this lesson

You will learn:

- how images are converted into digital data
- how audio sounds are converted into digital data.

Everything you store on a computer must be stored as digital data. In the last lesson you learned how numbers and text are stored as digital data. Images, sound and video are also converted into binary so that a computer can store and use them.

## Digital images

When you look at a photo on your computer screen it looks like the real world. In fact, it is an image made up of tiny squares called **pixels**. The word pixel is short for 'picture element'.

The pixels are organised in a grid of rows and columns. The picture is like a spreadsheet, but the cells contain colours not numbers. Each pixel stores a single colour.

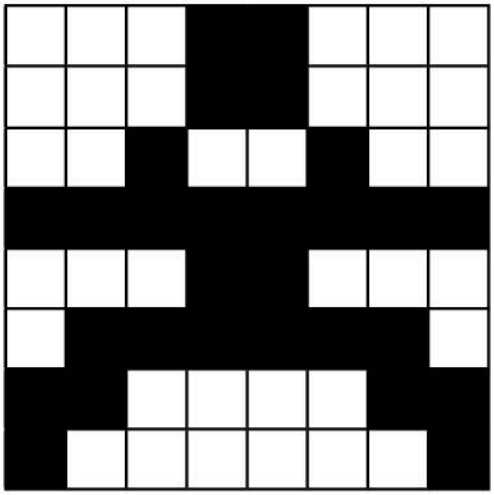
Pixels are so small that you cannot see the individual squares. Your brain blends the individual colours together to create a realistic image. An image created in this way is called a bitmap. Most images you see on computer screens are bitmaps.



Each colour the computer uses has its own binary code. The computer uses a colour code chart to convert the colour of each pixel into a binary code.

## Example 1: storing two colours

A simple image has been created on a square grid measuring 8 × 8 pixels. The image uses only two colours – black and white. All the information the computer needs to store the information about each pixel can be stored in a single bit.



| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |

If the pixel is white, the computer stores a 1 in the bit. If the pixel is black, the computer stores a 0. The computer can store all the information it needs about each line in this simple image using a single byte. The information looks like this:

11100111 11100111 11011011 00000000 11100111 10000001 00111100 01111110

### Activity

Draw a blank 8 × 8 pixel grid. Recreate the image stored in the digital data below. Work from the top left of the grid to the bottom right. 1 is a white square and 0 is black.

11111111 11100011 11011101 10101010 10111110 10100010 11011101 11100011

## Images with more colours

Most images use more than two colours. In Example 1, a single digit was used to store information about the colour of a pixel. A single digit can store just two values. To store more colours, a computer uses more bits.

## Example 2: storing more colours

In this example, the computer uses two bits to store the colour information about a single pixel. Using two bits means four colour codes can be used: 00, 01, 10 and 11. The codes used to store the colours in this example are: 00 black, 01 red, 10 blue and 11 white. The method is the same as in Example 1. Using more bits means you can use more colours in an image.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 11 | 11 | 00 | 00 | 00 | 11 | 11 | 11 | 00 | 00 | 00 | 11 | 11 |
| 11 | 00 | 01 | 01 | 01 | 00 | 11 | 00 | 01 | 01 | 01 | 00 | 11 |
| 00 | 01 | 01 | 01 | 01 | 01 | 00 | 01 | 11 | 11 | 01 | 01 | 00 |
| 00 | 01 | 01 | 01 | 01 | 01 | 01 | 11 | 10 | 01 | 01 | 01 | 00 |
| 00 | 01 | 01 | 01 | 01 | 01 | 01 | 11 | 01 | 01 | 01 | 01 | 00 |
| 00 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 00 |
| 11 | 00 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 00 | 11 |
| 11 | 11 | 00 | 01 | 01 | 01 | 01 | 01 | 01 | 01 | 00 | 11 | 11 |
| 11 | 11 | 11 | 00 | 01 | 01 | 01 | 01 | 01 | 00 | 11 | 11 | 11 |
| 11 | 11 | 11 | 11 | 00 | 01 | 01 | 01 | 00 | 11 | 11 | 11 | 11 |
| 11 | 11 | 11 | 11 | 11 | 00 | 01 | 00 | 11 | 11 | 11 | 11 | 11 |
| 11 | 11 | 11 | 11 | 11 | 11 | 00 | 11 | 11 | 11 | 11 | 11 | 11 |
| 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 | 11 |

## True colour

The examples in this lesson are simple images to explain how images can be stored as digital data. For most images, a computer uses more than two bits to store colour information.

For simple images like icons and emojis, the computer uses a byte (8 bits) to store colour information. A byte can store 256 different colours.

Photos need more than 256 colours to look realistic. A computer uses a method called **true colour** to store digital data about photos. True colour uses three bytes to store the information about a single pixel. True colour allows nearly 17 million colours to be used.

## Realistic images

Using more colours makes an image look realistic. This is called **colour depth**. Adding more colours increases the colour depth.

Another way to make images more realistic is to use more pixels. This is called **resolution**. Using more pixels to display an image gives high-resolution images.



Low resolution



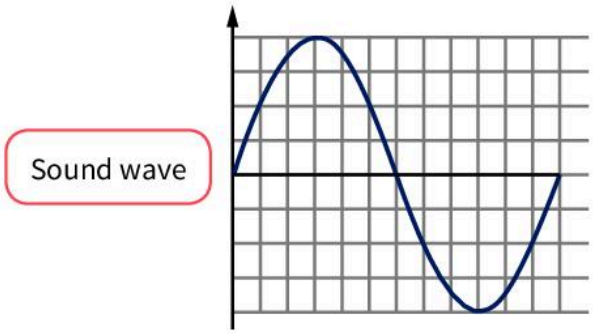
High resolution

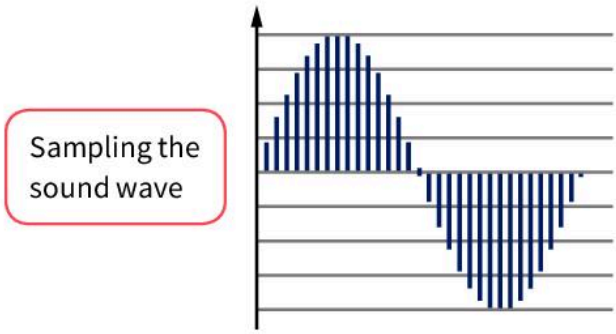
## Digital sound

If you press a piano key, a hammer strikes a string and the string vibrates. The vibration is a wave of sound that passes through the air to our ears. The sound wave is continuous and smooth.



A computer cannot save continuous data. The computer must break up continuous data into chunks that can be stored in bytes in its memory. This process is called **sampling**. A sample is a slice of the sound taken at a moment in time.




When sampling, the computer takes measurements of the continuous sound wave at regular intervals throughout the recording. The computer stores the measurements as digital data. The number of samples recorded per second is called the sampling rate. A piece of digital music is sampled around 44,000 times per second.

The sampling process never captures sound exactly. There are gaps between samples where sound is missed, but the gaps are so small that the digital sound appears continuous when we listen to it.

High-quality audio recordings have a high sampling rate. More samples are taken per second during the recording.

## Digital video

Digital video is created in the same way as audio. Sampling is used to capture slices of an event that happens continuously. The slices are called **frames**. When frames are displayed quickly one after the other, we see an image moving as it would in real life.

Earlier in this unit you created a flick book animation. The activity simulated the way that video is captured and displayed as digital data.

When a video is stored as a digital file, the images and audio are saved separately using the methods described in this lesson.

### Activity

Create your own two-colour picture in an 8 × 8 grid.

Convert each line in your grid into a byte: white = 1, black = 0.

Exchange your binary code with a partner.

Use the binary code to redraw your partner's image.

### Extra challenge

Create an information sheet that explains how sampling is used to capture and store audio in a digital data file. Search the web to find an image and more information about sampling.

### Test

1. Explain in your own words how pixels are used to make a digital image.
2. Describe how sampling is used to capture audio on a computer.
3. How are colour depth and resolution used to create high-quality images?
4. High-quality image files are larger than low-quality files. Why is that?

# Check what you know

## You have learned

- how text, images and audio can be stored as digital data by a computer
- how to convert between binary and decimal numbers
- how to add binary numbers.

Try the test and activity. They will help you to see how much you understand.

## Test

1 What are bits and bytes?

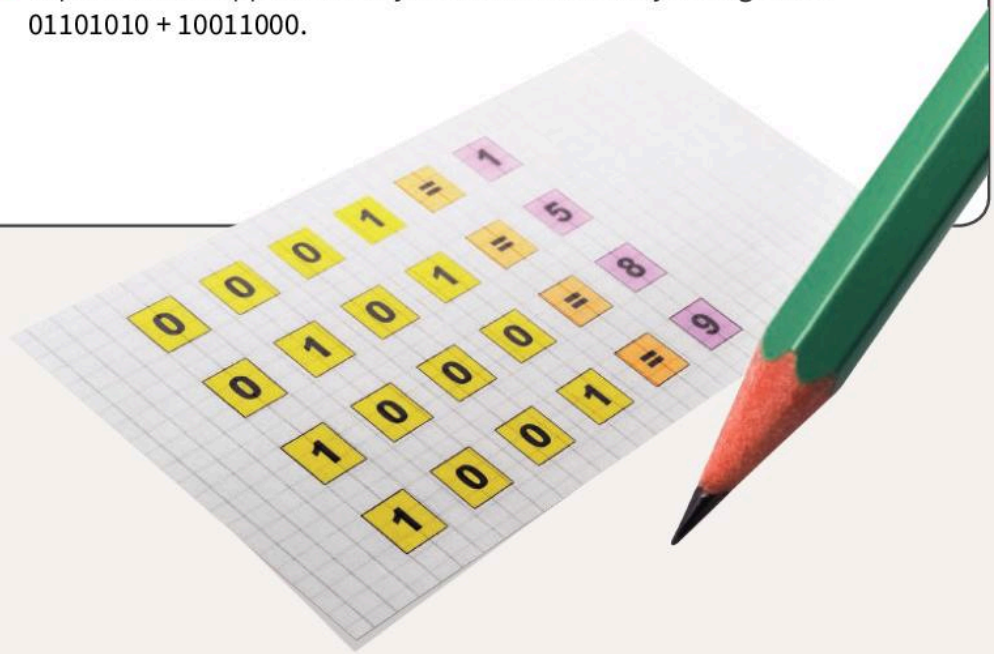
2 What is another name for the base 2 number system?

3 Show how to convert the decimal value 172 into binary.

4 Show how to add together the binary numbers 00011011 and 00101001. Do the same sum in decimal to check your answer.

5 Explain why the two 1s in this binary number do not have the same value: 00100010.

6 Explain what happens when you add these two bytes together: 01101010 + 10011000.



## Activity

Create a presentation to explain to other students how digital files are used to store text and images. Create the following slides for your presentation:

**Slide 1:** Explain that a computer stores everything as digital data.

**Slide 2:** Explain in your own words how a computer uses ASCII code to store letters in a digital file.

| Character | Binary Code |
|---|---|
| a | 01100001 |
| b | 01100010 |
| c | 01100011 |
| d | 01100100 |
| e | 01100101 |
| f | 01100110 |
| g | 01100111 |
| h | 01101000 |
| i | 01101001 |
| j | 01101010 |

| Character | Binary Code |
|---|---|
| 0 | 00110000 |
| 1 | 00110001 |
| 2 | 00110010 |
| 3 | 00110011 |
| 4 | 00110100 |
| 5 | 00110101 |
| 6 | 00110110 |
| 7 | 00110111 |
| 8 | 00111000 |
| 9 | 00111001 |

**Slide 3:** Explain how simple images can be stored as binary code in a digital file.

**Slide 4:** Explain how combining two or three bytes allows many characters or colours to be stored in a digital file. Use Unicode or true colour as an example. You can include information from your own web research on your slide.



### Self-evaluation

- I have answered test questions 1 and 2.
- I have created Slide 1 for my presentation.
- I have answered test questions 1–4.
- I have created Slides 1–3 for my presentation.
- I have answered all the test questions.
- I have created a presentation with all four slides.

Re-read any parts of the unit you feel unsure about. Try the test and activity again – can you do more this time?